Monkfish (*Lophius americanus*)

Summary Scores

- **Sensitivity:** [INSERT]
- **Exposure:** [INSERT]
- **Vulnerability:** [INSERT]

Sensitivity

Sensitivity Attributes	Range Score	Range LH DQ	Range Tally	Northern Score	Northern LH DQ	Northern Tally	Southern Score	Southern LH DQ	Southern Tally
Adult Mobility	1.72	2.75		1.72	2.75		1.72	2.75	
Complexity in Reproductive Strategy	2.18	2.45		2.16	2.45		2.2	2.45	
Dispersal of Early Life Stages	1.8	2.8		1.8	2.8		1.8	2.8	
Egg and Larval Survival and Settlement Requirements	1.68	2.2		1.64	2.2		1.72	2.2	
Historical Variability in Temperature	1.66	2.8		1.6	2.8		1.72	2.8	
Juvenile Habitat Requirements	1.22	3		1.2	3		1.24	3	
Other Stressors	1.52	2.83		1.48	2.75		1.56	2.9	
Population Growth Rate	2.6	2.61		2.6	2.66		2.6	2.56	
Prey Specificity	1.34	3		1.32	3		1.36	3	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.04	2.1		1.04	2.1		1.04	2.1	
Spawning Cycle	1.92	3		1.92	3		1.92	3	
Stock Size/Status	1.64	3		1.92	3		1.36	3	
Total Sensitivity | Certainty	**Low (1) | 1**			**Low (1) | 0.99**			**Low (1) | 0.98**		

Exposure

Exposure Factors	Range Mean	Range SD	Range Tally	Northern Mean	Northern SD	Northern Tally	Southern Mean	Southern SD	Southern Tally
Bottom Temperature	2.35	1.12		2.43	1.12		2.37	1.12	
Bottom Oxygen	2.71	1.01		3.1	0.88		2.47	1.02	
Bottom Salinity	2.75	0.99		2.18	1.06		3.16	0.69	
Bottom Aragonite Solubility	1.32	0.78		1.14	0.39		1.42	0.93	
Small zooplankton integrated biomass	1.3	0.35		1.28	0.24		1.34	0.42	
Medium zooplankton integrated biomass	1.38	0.33		1.46	0.24		1.32	0.39	
Large zooplankton integrated biomass	1.24	0.32		1.35	0.35		1.12	0.24	
Total Exposure	**Moderate (2.02)**	**0.95**		**Moderate (1.99)**	**0.96**		**Moderate (2.08)**	**0.95**	

Vulnerability

Vulnerability	Range Mean	Range SD	Range Tally	Northern Mean	Northern SD	Northern Tally	Southern Mean	Southern SD	Southern Tally
Total Vulnerability	**Low (2.02)**	**0.95**		**Low (1.99)**	**0.96**		**Low (2.08)**	**0.95**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Monkfish. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 6 and Figure 5). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	[INSERT]	[INSERT]	[INSERT]
Exposure	[INSERT]	[INSERT]	[INSERT]
Vulnerability	[INSERT]	[INSERT]	[INSERT]
Directional Effect	[INSERT]	[INSERT]	[INSERT]

Stocks

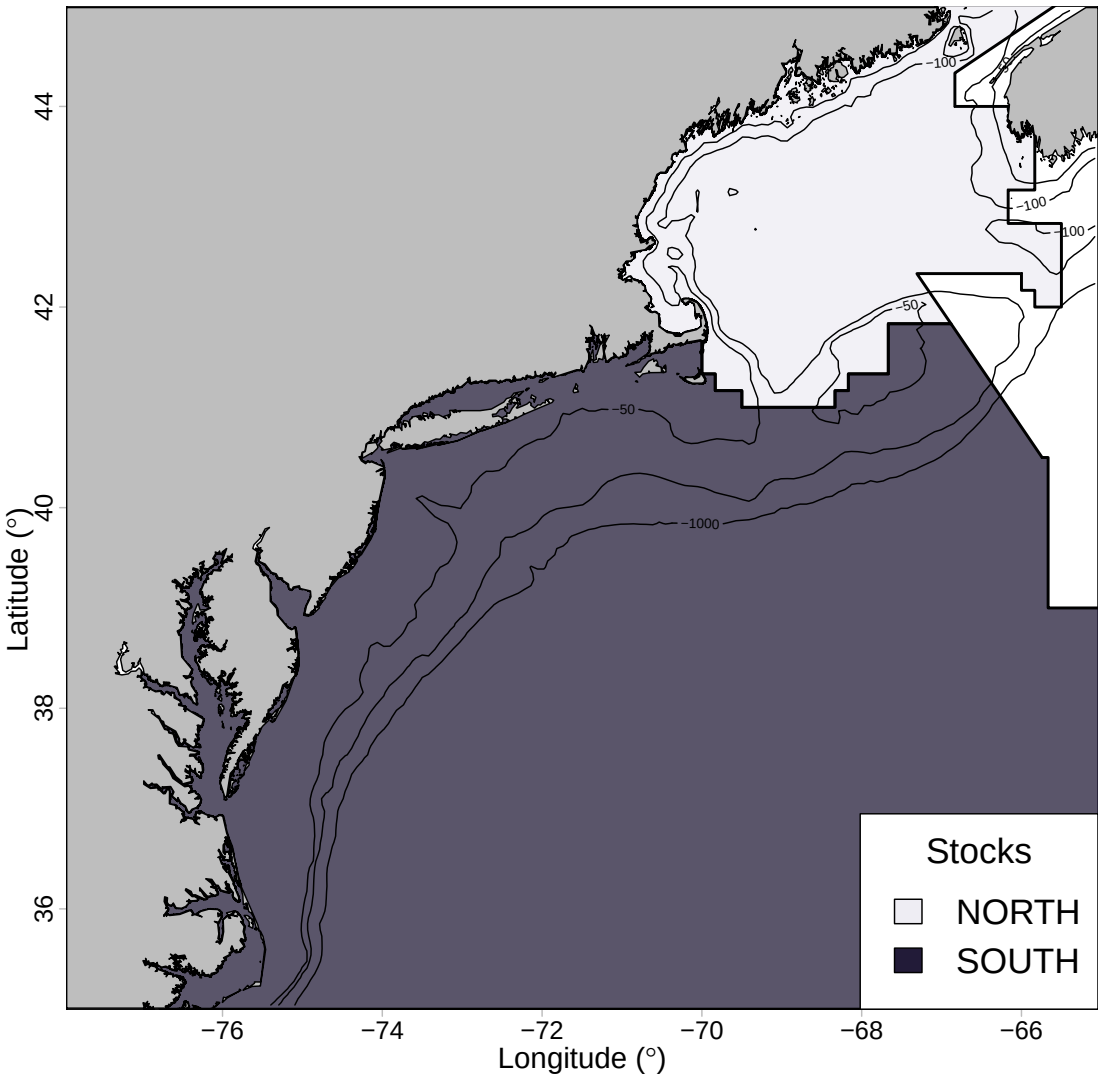


Figure 2: Monkfish Stocks along the Northeast Shelf

Score Descriptions:

Sensitivity:

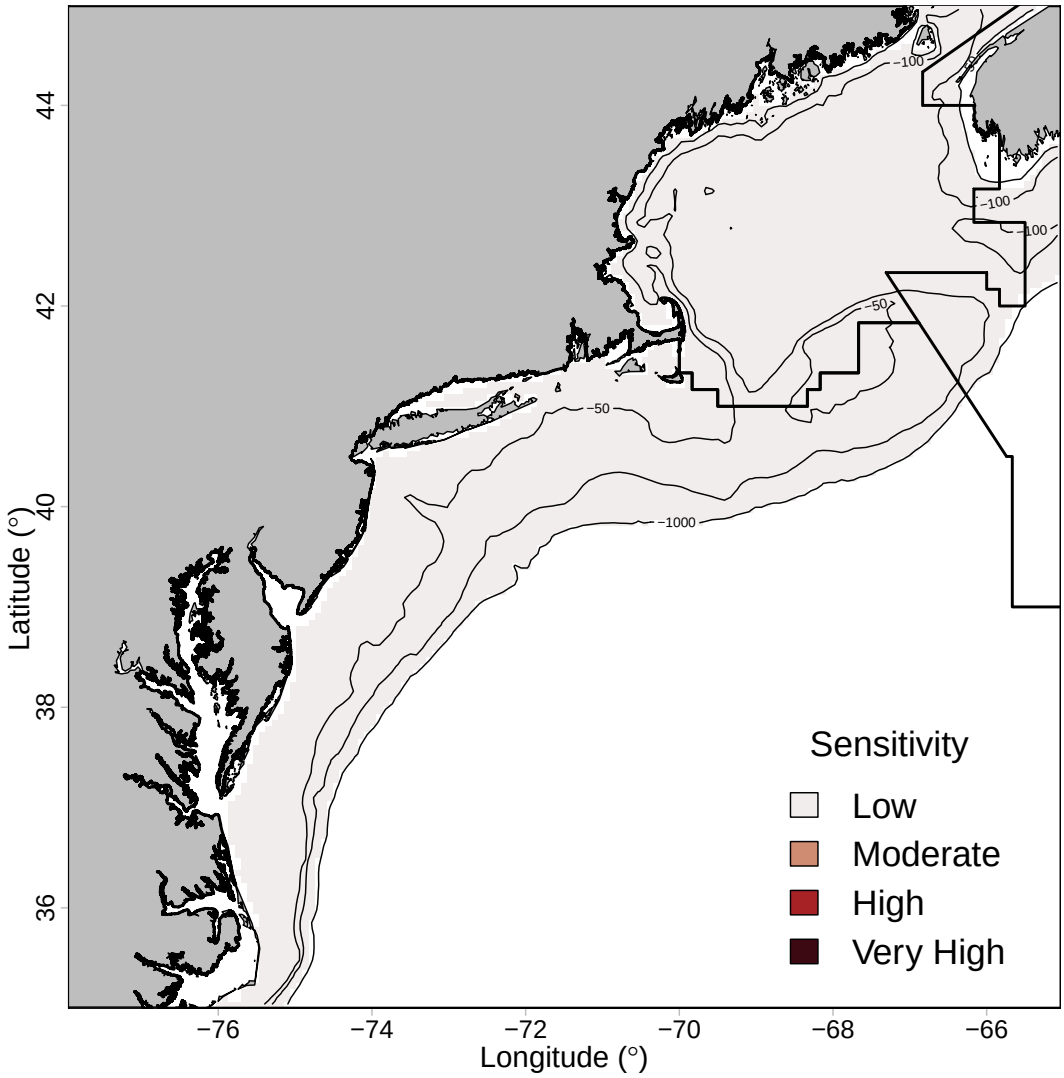


Figure 3: Monkfish Sensitivity on the Northeast Shelf

Both the **Northern** and **Southern** Stocks of Monkfish received a **Low** sensitivity score. Certainties were 99% and 98% respectively (Figure 1). For both stocks, the only attribute that received a score greater than 2.5 was Population Growth Rate (2.6 for both stocks) (Figure 1). Monkfish mature slowly, reaching maturity at 4-5 years (Richards et al. 2008). Maximum ages range from 7 years for males to 14 years for females (Northeast Fisheries Science Center 2010).

Monkfish also received a Low sensitivity score in the 2016 Climate Vulnerability Assessment for the Northeast Shelf. Like in this assessment, this result was driven by Population Growth Rate (Hare et al. 2016).

Exposure:

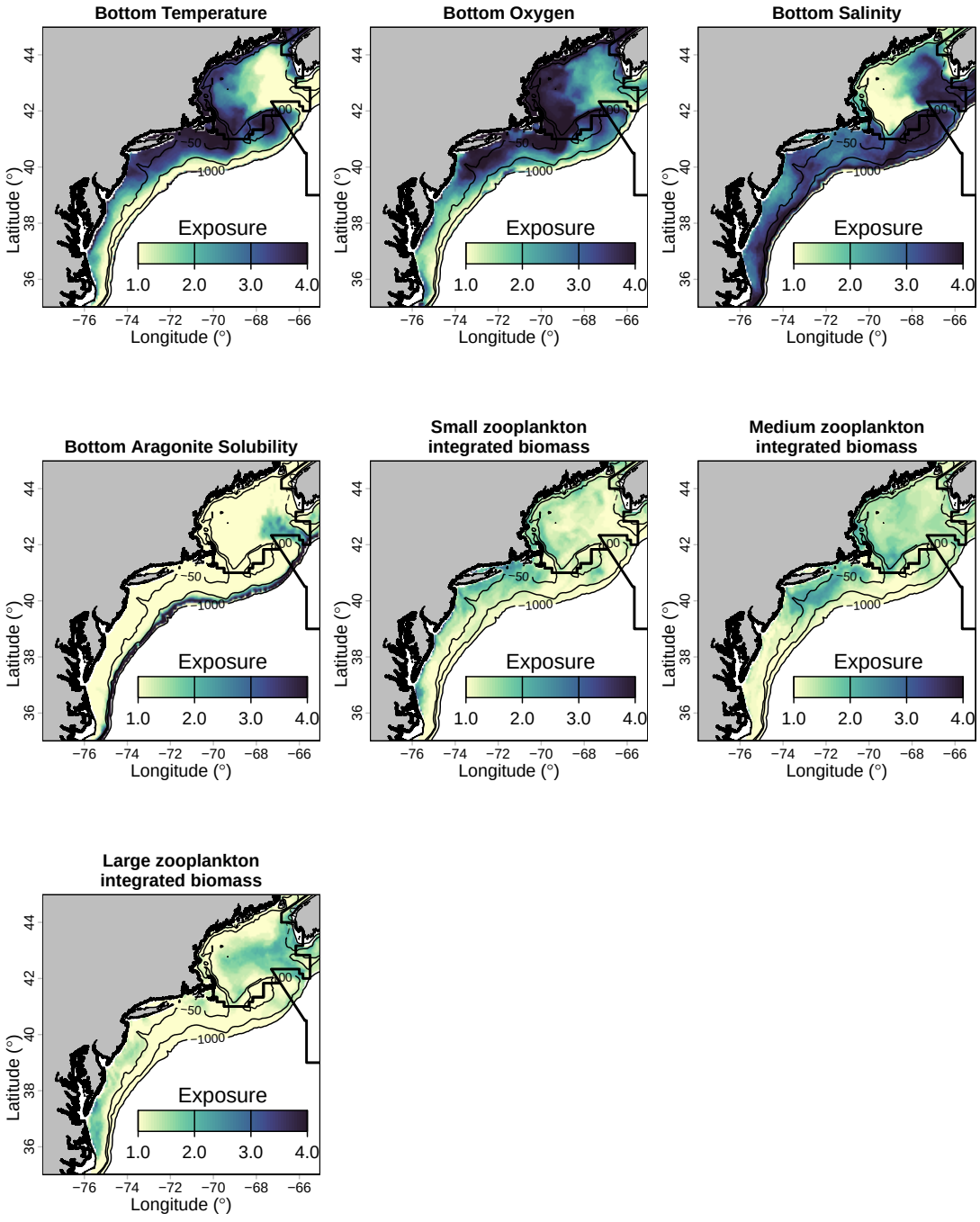


Figure 4: Maps of each Exposure Factor for Monkfish. Exposure was calculated between 2014-2023 and 2025-2035

Mean Exposure for Monkfish was **Moderate** throughout its range (2.02 +/- 0.95). Scores were also **Moderate** when averaged across both the **Northern** and **Southern** stocks (1.99 +/- 0.96 and 2.08 +/-

0.95, respectively) (Figure 1). Exposure was greatest (Very High) over Georges Bank and Moderate in Southern New England and along the coast of the Gulf of Maine (Figure 5). This was driven by bottom temperatures, oxygen, and salinity, as well as zooplankton biomass exposure. Exposure was also moderate on the shelf break due to high exposure to changes in bottom aragonite solubility (Figure 4).

In the previous Climate Vulnerability Assessment for the Northeast Shelf, monkfish received a High Exposure score. This previous score was driven by sea surface temperature, ocean acidification, and currents (Hare et al. 2016).

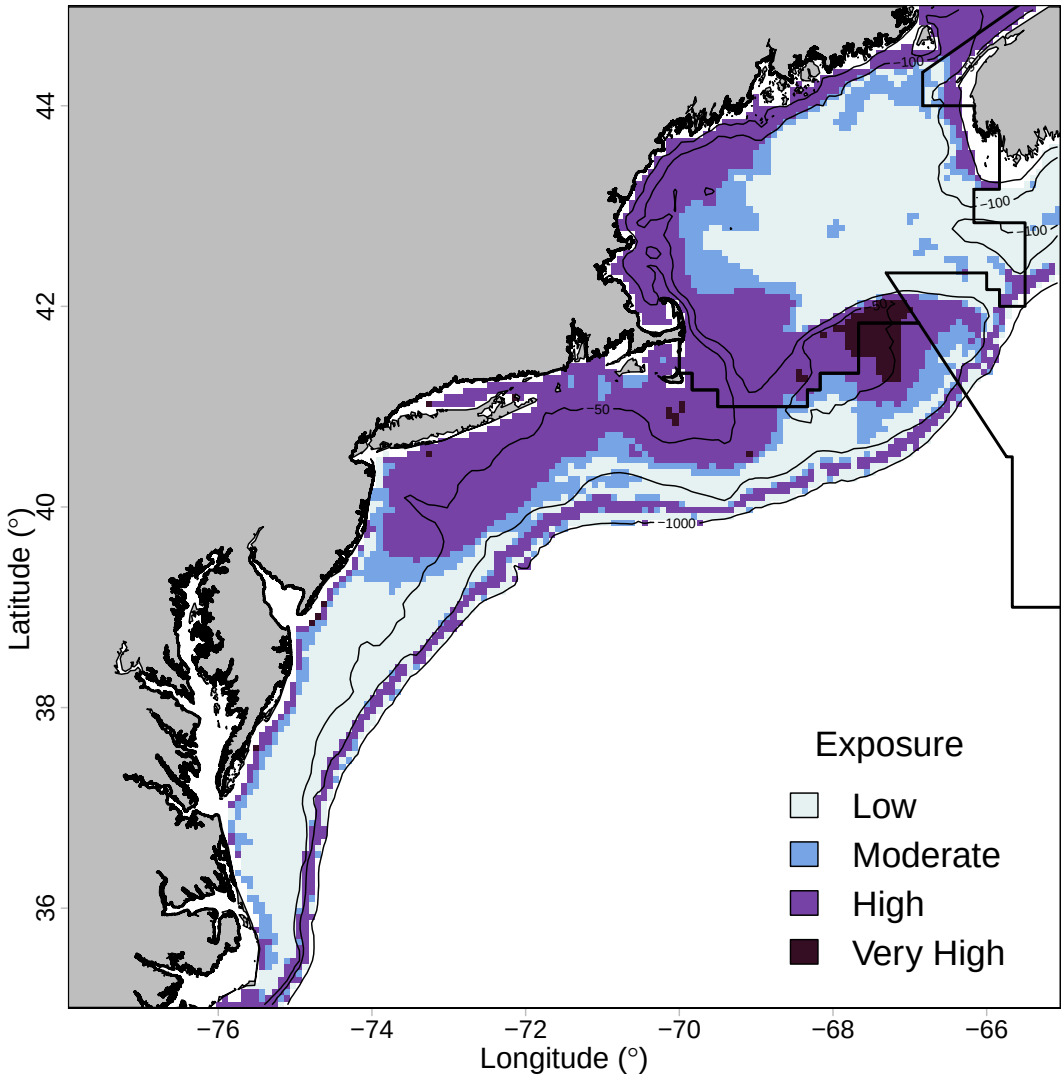


Figure 5: Total Exposure on the Northeast Shelf for Monkfish. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 4)

Vulnerability:

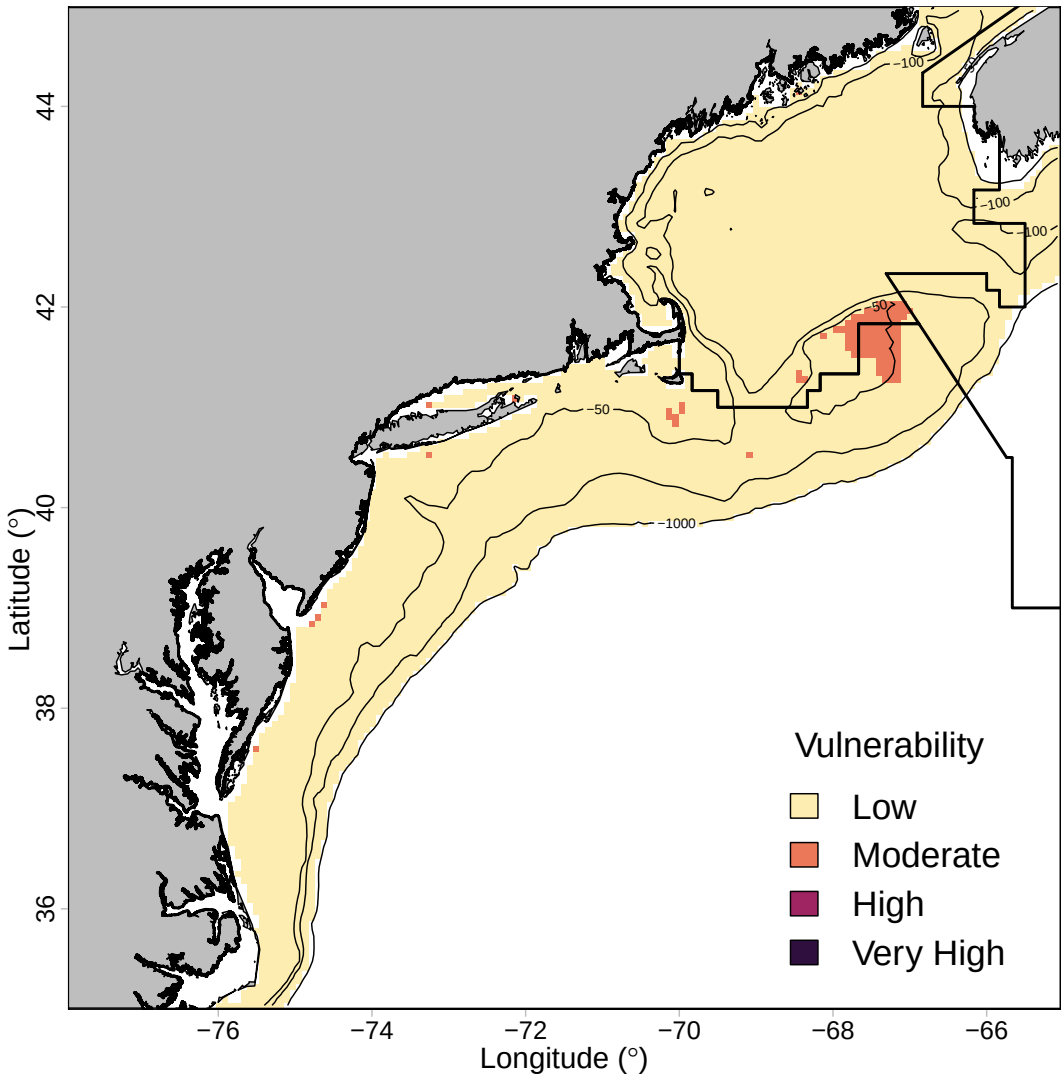


Figure 6: Vulnerability on the Northeast Shelf for Monkfish. Vulnerability is the product of sensitivity and exposure (see Figure 5 and Figure 3)

Mean Exposure for Monkfish was **Moderate** throughout its range (2.02 +/- 0.95). Scores were also **Moderate** when averaged across both the **Northern** and **Southern** stocks (1.99 +/- 0.96 and 2.08 +/- 0.95, respectively) (Figure 1). Exposure was greatest (Very High) over Georges Bank and Moderate in Southern New England and along the coast of the Gulf of Maine (Figure 5). This was driven by bottom temperatures, oxygen, and salinity, as well as zooplankton biomass exposure. Exposure was also moderate on the shelf break due to high exposure to changes in bottom aragonite solubility (Figure 4).

In the previous Climate Vulnerability Assessment for the Northeast Shelf, monkfish received a High Exposure score. This previous score was driven by sea surface temperature, ocean acidification, and currents (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

88% of data quality scores were greater than or equal to 2, indicating strong data quality. No data gaps (mean attribute data quality score < 2) were identified (Figure 1).

Model Confidence Score:

The monkfish ensemble model received a 2.4 (+/- 0.4) model confidence score, indicating moderate confidence in the model. The final model met experts' expectations, representing the known distribution and drivers well. The ensemble model AUC score was 0.94, indicating that the model had excellent predictive performance.

Additional Metrics

Distribution Shifts:

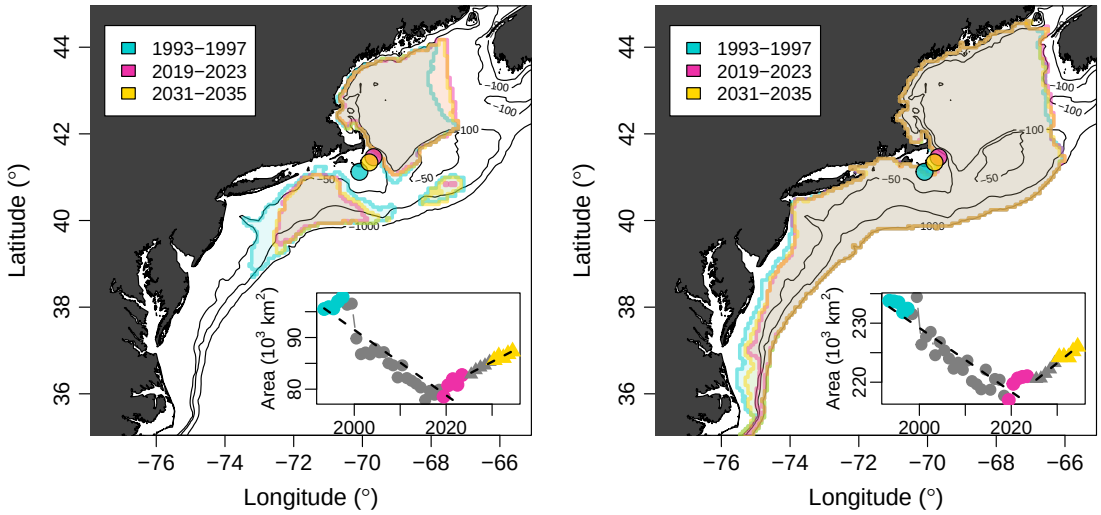


Figure 7: Distribution shifts on the Northeast Shelf for Monkfish. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Monkfish lost significant amounts of habitat between 1993 - 2023, particularly off of southern New England and at the southern edge of their range. Monkfish have gained habitat in the Gulf of Maine and total habitat area has increased in recent years (Figure 7). The weighted centroid has shifted 51 km northeast between 1993 and 2023.

Under near future ocean conditions, the recent habitat growth trend is expected to continue. Monkfish habitat is predicted to expand south of Georges Bank and at the southern edge of their range (Figure 7). The weighted centroid is expected to shift 17 km southwest, back towards its 1993 location.

In the previous Assessment, monkfish received a High Distributional Vulnerability rank because of their habitat preferences (generalists) and seasonal migrations (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	96,668	233,269	6.23	8.02
2019-2023	50.9 km NE	81,028 (-15,640)	220,138 (-13,131)	5.23 (-1.00)	7.22 (-0.80)
2031-2035	16.6 km SW	86,432 (+5,404)	224,828 (+4,689)	5.80 (+0.56)	7.70 (+0.48)

Directional Effect:

Climate change is predicted to **negatively** impact both the **northern** and **southern** stocks of Monkfish. Certainties were 94% and 87% respectively. Monkfish are most abundant on Georges Bank when bottom temperatures are between 5-14 C, suggesting that warming temperatures would decrease the amount of preferred habitat (Siemann et al. 2018). Experts in the previous Assessment predicted that climate change would have a neutral impact on Monkfish with a high degree of uncertainty (Hare et al. 2016).

Climate Effects on Abundance and Distribution

In Icelandic waters, monkfish distribution changes and increased stock size have been correlated with rising water temperatures and expanding suitable habitat (Solmundsson et al. 2007). Monkfish movement and relative abundance on Georges Bank is driven by seasonal changes in bottom temperature. Their distribution may shift in response to future changes in thermal habitat, highlighting the need for further management strategies (Siemann et al. 2018). Maximum size of monkfish in the Northwest Atlantic has decreased, and is related to seasonality and bottom temperature (Charbonneau et al. 2020).

Life History Synopsis

Monkfish are a bottom dwelling fish ranging from the northern Gulf of St. Lawrence to Cape Hatteras. The commercial fishery operates from Maine to North Carolina (Steimle et al. 1999). Spawning takes place from spring through early autumn, progressing from south to north over a 6 month period (Richards et al. 2008). Pelagic larvae hatch from egg veils (up 12 m long and 1.5 m wide) 1-3 weeks after being laid, depending on temperature. Larvae remain in the egg veil 2-3 days after hatching. Monkfish larvae feed on zooplankton, including copepods, crustacean larvae, and chaetognaths (Bigelow and Schroeder 1953). Larval to juvenile transition occurs at 5-10 cm TL (total length) when the elongate fins and body gradually assume the adult form (Northeast Fisheries Science Center 2010). Once larvae transition to juveniles, they move to the benthos. Juveniles and adults settle on mud, gravelly sand, algae, and rock substrates (Steimle et al. 1999). Once moved to the bottom, juveniles can eat sand lance, but invertebrates such as red shrimp

and squid make up the majority of their diet. The amount of invertebrates eaten by juveniles decreases as they get older (Steimle et al. 1999). Monkfish are ambush predators and opportunistic feeders; prey found in their stomachs include a variety of benthic and pelagic species. Monkfish are also cannibalistic and can eat sea birds and diving ducks. Adults spend most of their time resting on the bottom, but have been reported at the surface after a storm event, preying on seabirds (Richards et al. 2008). Large monkfish have few predators, but smaller monkfish are common prey for swordfish, sharks, and skates (Steimle et al. 1999). Monkfish move by slow swimming or by using their sturdy pectoral fins to "walk." Despite slow speeds, juveniles and adults migrate in response to thermal conditions, spawning, or food availability (Steimle et al. 1999). Monkfish were found on Georges Bank in summer and early fall, and found in deeper waters/on the edge of the bank in winter months, suggesting that movements may be influenced by temperature (Rountree et al. 2006). Age at maturity is 4-5 years, longer for females and for the Mid-Atlantic stock (Richards et al. 2008). Maximum age is listed at 7 years for males and 12-14 years for females (Northeast Fisheries Science Center 2010). Monkfish in the Northwest Atlantic are divided into two stocks: the Gulf of Maine/Northern Georges Bank and Southern Georges Bank/Mid-Atlantic. Both stocks do not have stock assessments that provide indices on overfishing and overfished status, although an index based assessment is used for catch advice (Deroba 2022).

Supplemental Scores

Exposure Timeseries

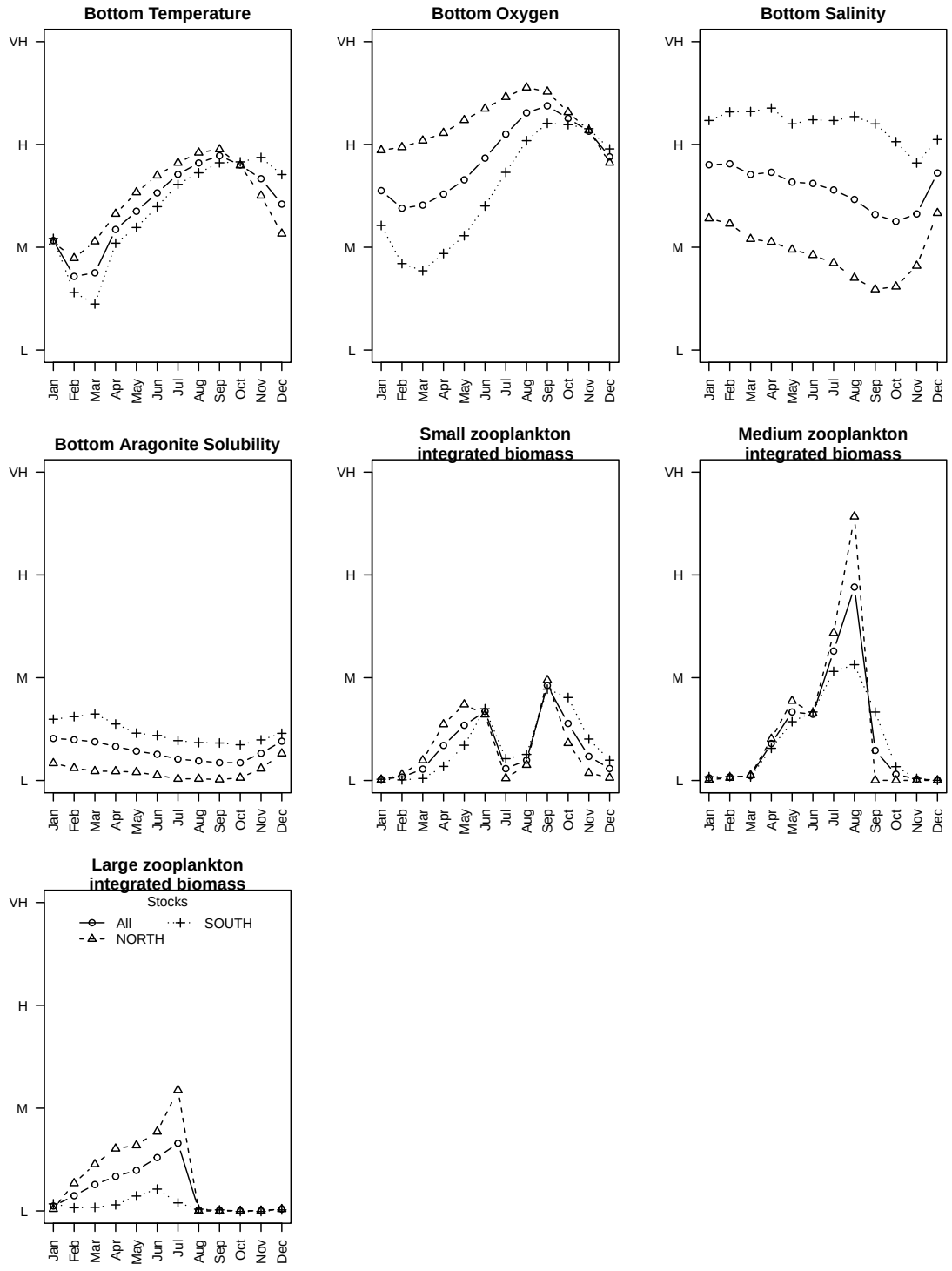


Figure 8: Monthly Timeseries of each Exposure Factor for Monkfish. Exposure was calculated between 2014-2023 and 2025-2035

Within the Northern and Southern stocks of Monkfish, monthly exposures ranged from Low to High (Figure 9). In the Northern Stock, Exposure was Moderate from May to November and increased to High in August. In the Southern Stock, Exposure was High from August through October. These patterns were driven by increases in bottom temperature and oxygen exposure as the year progressed, and sharp increases in medium zooplankton biomass in late summer (Figure 8).

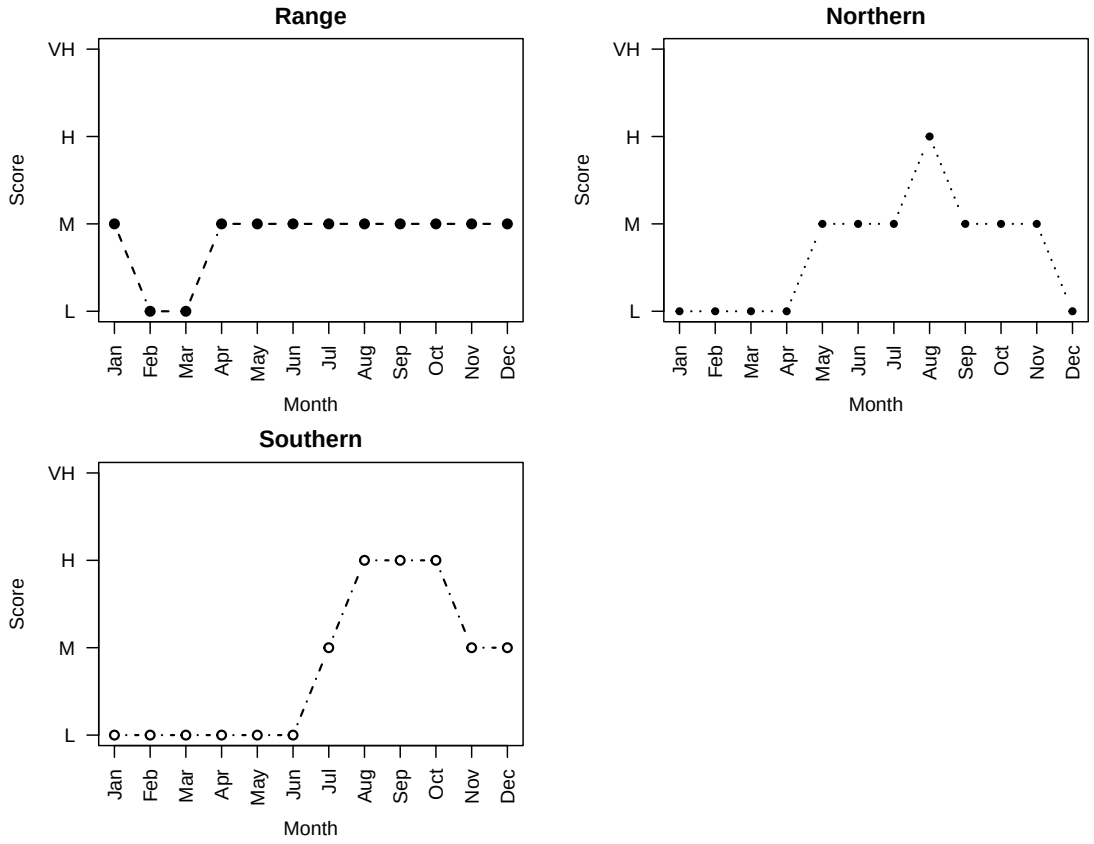


Figure 9: Timeseries of Total Exposure for Monkfish

Exposure due to Significant Factors

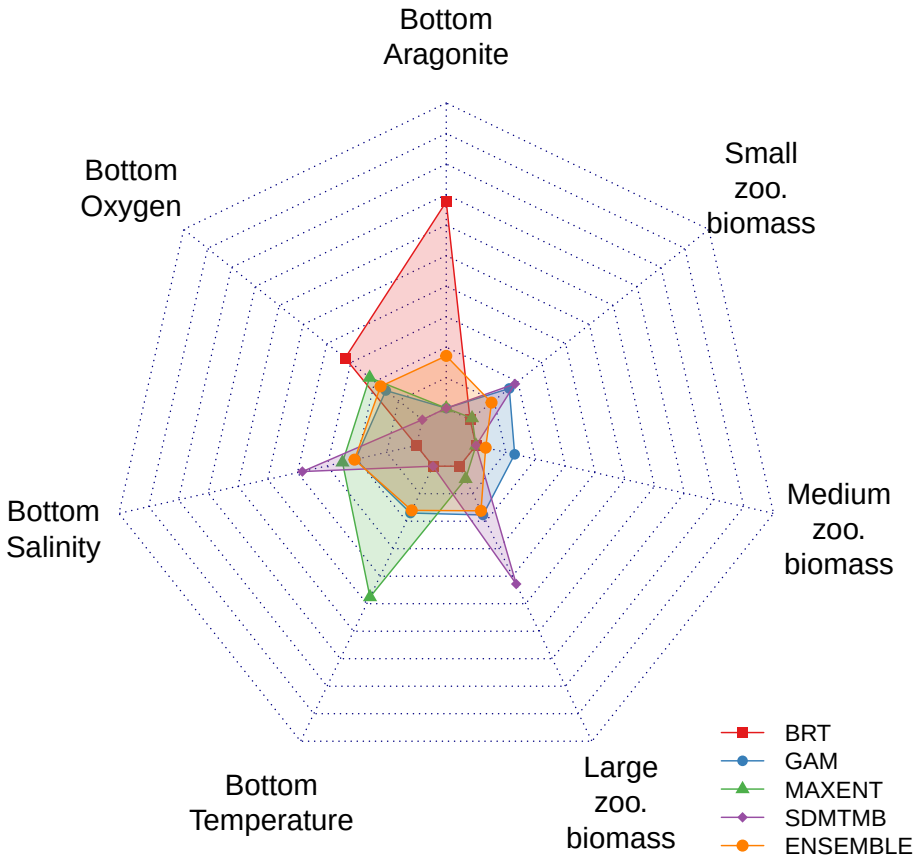


Figure 10: Dynamic Factors driving Monkfish distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Several factors were considered important dynamic drivers of Monkfish distributions: bottom temperature, oxygen, salinity, and large zooplankton biomass (Figure 10). Adult monkfish are demersal (Richards et al. 2008) and movement patterns have been associated with bottom temperatures (Siemann et al. 2018). Monkfish are generalist predators but larvae prey on zooplankton (Bigelow and Schroeder 1953). When only these factors were used to calculate Exposure, scores remained Moderate both throughout the species range and within the stocks (Figure 11, Figure 12).

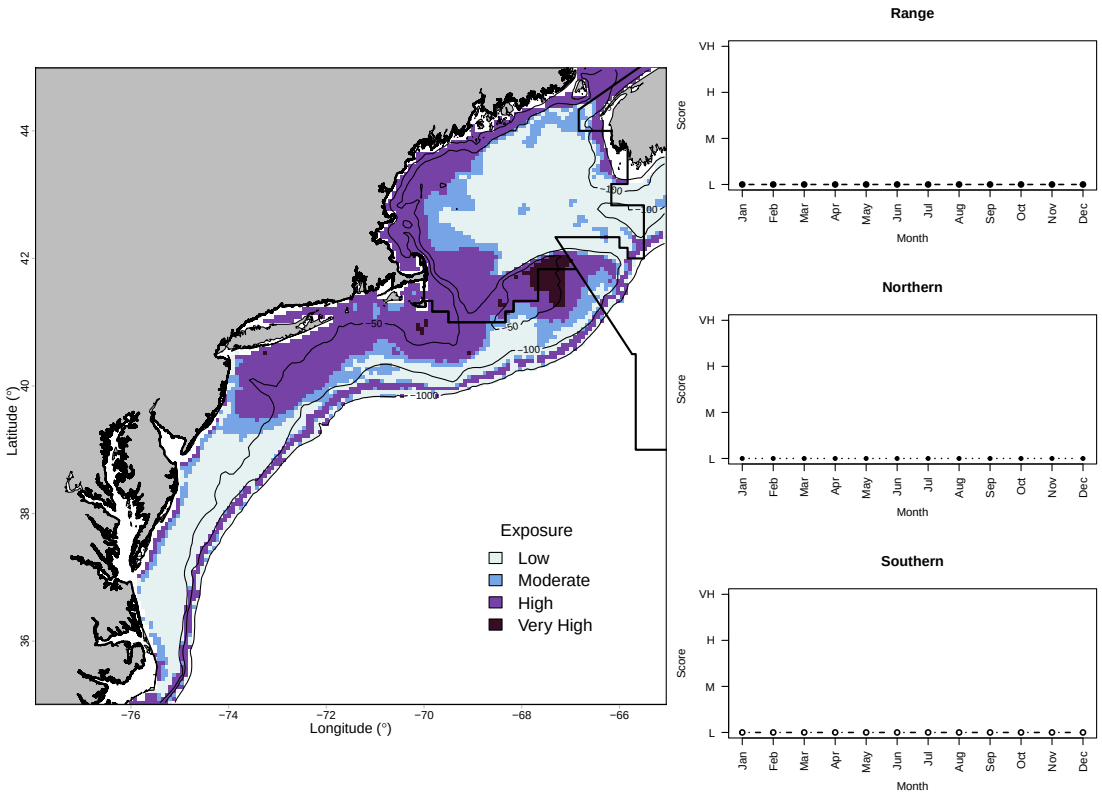
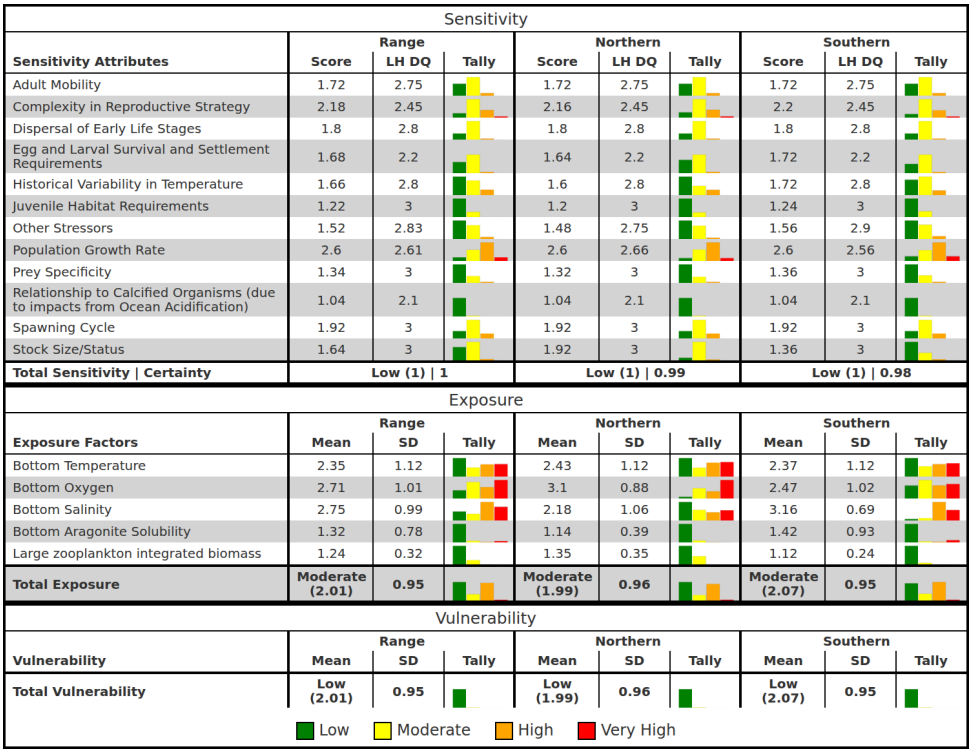


Figure 11: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Monkfish distributions



Sensitivity

Sensitivity Attributes	Range Score	Range LH DQ	Range Tally	Northern Score	Northern LH DQ	Northern Tally	Southern Score	Southern LH DQ	Southern Tally
Adult Mobility	1.72	2.75		1.72	2.75		1.72	2.75	
Complexity in Reproductive Strategy	2.18	2.45		2.16	2.45		2.2	2.45	
Dispersal of Early Life Stages	1.8	2.8		1.8	2.8		1.8	2.8	
Egg and Larval Survival and Settlement Requirements	1.68	2.2		1.64	2.2		1.72	2.2	
Historical Variability in Temperature	1.66	2.8		1.6	2.8		1.72	2.8	
Juvenile Habitat Requirements	1.22	3		1.2	3		1.24	3	
Other Stressors	1.52	2.83		1.48	2.75		1.56	2.9	
Population Growth Rate	2.6	2.61		2.6	2.66		2.6	2.56	
Prey Specificity	1.34	3		1.32	3		1.36	3	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.04	2.1		1.04	2.1		1.04	2.1	
Spawning Cycle	1.92	3		1.92	3		1.92	3	
Stock Size/Status	1.64	3		1.92	3		1.36	3	
Total Sensitivity | Certainty	**Low (1) | 1**			**Low (1) | 0.99**			**Low (1) | 0.98**		

Exposure

Exposure Factors	Range Mean	Range SD	Range Tally	Northern Mean	Northern SD	Northern Tally	Southern Mean	Southern SD	Southern Tally
Bottom Temperature	2.35	1.12		2.43	1.12		2.37	1.12	
Bottom Oxygen	2.71	1.01		3.1	0.88		2.47	1.02	
Bottom Salinity	2.75	0.99		2.18	1.06		3.16	0.69	
Bottom Aragonite Solubility	1.32	0.78		1.14	0.39		1.42	0.93	
Large zooplankton integrated biomass	1.24	0.32		1.35	0.35		1.12	0.24	
Total Exposure	**Moderate (2.01)**	**0.95**		**Moderate (1.99)**	**0.96**		**Moderate (2.07)**	**0.95**	

Vulnerability

Vulnerability	Range Mean	Range SD	Range Tally	Northern Mean	Northern SD	Northern Tally	Southern Mean	Southern SD	Southern Tally
Total Vulnerability	**Low (2.01)**	**0.95**		**Low (1.99)**	**0.96**		**Low (2.07)**	**0.95**	

Low Moderate High Very High

Figure 12: As in Figure 1 for Monkfish, but with only significant Exposure factors and the resulting Vulnerability Scores.

References

Bigelow, H. B., and W. C. Schroeder. 1953. *Fishes of the Gulf of Maine*. Vol. 53. U.s. Fish and Wildlife Service Fishery Bulletin. U.S. Government Printing Office.

Charbonneau, J. A., D. M. Keith, M. A. MacNeil, J. A. Sameoto, and J. A. Hutchings. 2020. "Pervasive Declines in Monkfish (Lophius americanus) Size Structure Throughout the Northwest Atlantic." *Fisheries Research* 230: 105633. https://doi.org/10.1016/j.fishres.2020.105633.

Deroba, J. 2022. *Draft 2022 Monkfish Management Track Assessment Report*. National Oceanic; Atmospheric Administration (NOAA). https://apps-st.fisheries.noaa.gov/sis/docServlet?fileAction=download&fileId=8456.

Hare, Jonathan A, Wendy E Morrison, Mark W Nelson, et al. 2016. "A Vulnerability Assessment of Fish and Invertebrates to Climate Change on the Northeast US Continental Shelf." *PloS One* 11 (2): e0146756.

Northeast Fisheries Science Center. 2010. *50th Northeast Regional Stock Assessment Workshop (50th SAW)*

Assessment Report. Nos. 10-17. Northeast Fisheries Science Center Reference Document. National Marine Fisheries Service. https://repository.library.noaa.gov/view/noaa/3744.

Richards, R. A., P. Nitschke, and K. Sosebee. 2008. "Population Biology of Monkfish Lophius americanus." *ICES Journal of Marine Science* 65 (7): 1291–305. https://doi.org/10.1093/icesjms/fsn131.

Rountree, R. A., J. P. Gr"oger, and D. Martins. 2006. "Extraction of Daily Activity Pattern and Vertical Migration Behavior from the Benthic Fish, Lophius americanus, Based on Depth Analysis from Data Storage Tags." *ICES Annual Science Conference 2006*, Theme session q. https://doi.org/10.17895/ices.pub.25259230.v1.

Siemann, L. A., C. J. Huntsberger, J. S. Leavitt, and R. J. Smolowitz. 2018. "Summering on the Bank: Seasonal Distribution and Abundance of Monkfish on Georges Bank." *PLOS ONE* 13 (11): e0206829. https://doi.org/10.1371/journal.pone.0206829.

Solmundsson, J., E. Jonsson, and H. Bjornsson. 2007. *Recent Changes in the Distribution and Abundance of Monkfish (Lophius piscatorius) in Icelandic Waters*. ICES Document CM 2007/B:02. International Council for the Exploration of the Sea.

Steimle, F. W., W. W. Morse, and D. Johnson. 1999. *Essential Fish Habitat Source Document: Goosefish, Lophius americanus, Life History and Habitat Characteristics*. No. 127. NOAA Technical Memorandum NMFS-NE. National Oceanic; Atmospheric Administration (NOAA).